

AWS Relational Database Service:-

What is AWS RDS?

• Traditionally, to create a database, you:

1. install an operating system
2. install a database (MySQL, Oracle, etc.) on it
3. maintain everything yourself (backup, updates, replication).

• with AWS RDS:-

• Amazon manages the database server for you.

• You just launch an RDS instance and use it.

• Backups, patching, and maintenance are done automatically.

Definition:-

Amazon RDS is a cloud service that makes it easy to set up, run, and scale a relational database.

it:

provides fast performance

- Has high availability,
- Offers security and compatibility.
- Supports 6 database engine;
 - Amazon Aurora
 - PostgreSQL
 - MySQL
 - MariaDB
 - Oracle Database
 - Microsoft SQL Server.

Benefits:-

- Easy to administer - create database in minutes using:
 - AWS console
 - AWS CLI
 - AWS API
- High scalable - increase compute/storage without downtime.
- Available + Durable - multi-AZ deployment for failover.
- Secure - Use IAM to control access.
- Cost-effective - pay only for what you use.

Free tier Pricing.

750 hrs / month of db-t2.micro (single-AZ) for MySQL, MariaDB, PostgreSQL, Oracle BYOL, or SQL Server Express.

- 20 GB SSD storage
- 20 GB storage for backups + snapshots.

Key - Features:-

1. Replication

- create Read replicas for read-heavy workloads.
- multi-AZ for automatic failover with synchronous replication.

2. Storage Type:-

- General purpose SSD (default, balanced).
- provisioned IOPS SSD (high-performance).
- magnetic (low-cost, slower).

3. Monitoring:-

integrated with Amazon CloudWatch

4. Patching:-

Automatic database engine updates.

5. Backups:-

- Automatic backup with Point-in-time recovery.
- manual snapshots available.

6. Pay-as-you-go

- No upfront cost, pay for usage

RDS (Relational Database Service) components

1. DB instances:-

- The main building block.
- isolated environment with its own compute, memory, and storage.
- can hold multiple databases.
- size 5GB to 6TB

2. Security Groups:-

- Control who can access the DB instance (by IP or EC2).

3. DB Parameter Groups:-

- store database engine configuration values
- Default group used if none assigned.

4. DB Option Groups:-

- Enable special database features/tools provided by the engine.

DevOps Roles & Responsibilities for Database.

1. create and manage apps Databases for different environments (dev, QA, Prod)
2. monitor database health and performance
3. create database and tables as per application needs.
4. configure database setting in apps or servers.
5. Handle backup, updates, and Patching.
6. Support DB, QA and Dev teams with their database needs.
7. Manage multiple DB instances for different environments.
8. Create read replicas for better performance and availability.